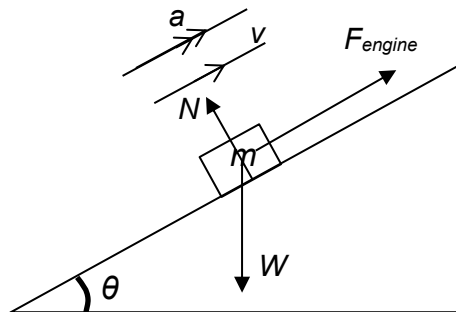


7

C

Consider FBD for car,



For car to experience forward acceleration of a at that instant, there must be a net force acting in the forward direction. Considering forces acting along the slope,

$$F_{net} = F_{engine} - W_{along_slope}$$

$$ma = F_{engine} - mg \sin(\theta)$$

$$F_{engine} = ma + mg \sin(\theta)$$

Power required by engine

$$P = F_{engine} v$$

$$P = (ma + mg \sin(\theta))v$$

8

C

Loss in GPE = Gain in KE